

SEMESTER END EXAMINATIONS - FEBRUARY 2024

Program	: B.E :- Information Science and Engineering	Semester	: V
Course Name	: Computer Networks	Max. Marks	: 100
Course Code	: IS53	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Enumerate the functions of each layer of TCP/IP model (protocol suite) with its layered diagram. What do you mean by encapsulation and decapsulation in the context of data communication? Why are they required? CO1 (08)
 - Compare star topology with mesh topology with an example network of 5 nodes. CO1 (06)
 - Draw a graph of the NRZ-I, Manchester and Differential Manchester line coding schemes for the data **010110111**. CO1 (06)
- What is transmission impairment? Briefly discuss the causes for transmission impairment. CO1 (10)
 - A periodic composite signal has a bandwidth of 300 kHz, with a middle frequency of 120kHz and peak amplitude of 10 V. The two extreme frequencies have amplitude of 3 V. Represent the signal in time and frequency domain. CO1 (04)
 - Compare TCP/IP protocol stack with OSI model. Mention the reasons for lack of OSI model success. CO1 (06)

UNIT - II

- Given the dataword 101001111 and the divisor 10111, show the generation of the CRC codeword at the sender site (using Binary division). CO2 (08)
 - Differentiate between pure ALOHA and Slotted ALOHA. CO2 (08)
 - Apply the even parity generator, find the parity bit for each of the following data units.
i) 1001011 ii) 0001100 iii) 10000000 iv) 1110111. CO2 (04)
- A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system produce
i) 1000 frames per second?
ii) 500 frames per second?
iii) 250 frames per second? CO2 (08)
 - Differentiate between CSMA and CSMA/CD. CO2 (06)
 - Illustrate Forward Error Correction using Hamming Distance with an example. CO2 (06)

UNIT - III

5. a) What are the restrictions imposed on classless address blocks? An Organization is granted the block 211.17.180.0/24. The administrator wants to create 32 subnets. CO3 (08)
- (i) Find the subnet mask
 - (ii) Find the number of addresses in each subnet
 - (iii) Find the first and last address of subnet 1
 - (iv) Find the first and last address of subnet 32
 - (v) Find the network address of subnet 3.
- b) Illustrate two node instability problem in Distance Vector Routing with an example. Discuss the solutions to solve the same. CO3 (08)
- c) What is fragmentation in Internet protocol? Discuss the fields related to fragmentation. CO3 (04)
6. a) Illustrate with diagram, the procedure for network address translation (NAT) using one IP address. CO3 (08)
- b) Discuss the different open-loop congestion control techniques. CO3 (05)
- c) An IPV4 datagram has arrived with the following header information (in hexadecimal values) - CO3 (04)
- 45 00 00 54 00 03 58 50 20 06 00 00 7C 4E 03 02 B4 0E 0F 02**
- Answer the following questions for the above IPV4 datagram:
- (i) Are there any options?
 - (ii) Is the packet fragmented?
 - (iii) What is the size of the data?
 - (iv) How many hops can this packet travel before being dropped?
- d) Differentiate between connection oriented service and connectionless service. CO3 (03)

UNIT- IV

7. a) Illustrate IPV6 datagram format and mention the significance of each field. CO4 (08)
- b) Compare connectionless and connection oriented protocol. CO4 (08)
- c) Illustrate selective-repeat request protocol in sender and receiver window with an example. CO4 (04)
8. a) Briefly explain user datagram protocol with supporting diagram. CO4 (08)
- b) Discuss stop and wait protocol with an example. CO4 (08)
- c) Explain processes to process communication and addressing in transport layer services. CO4 (04)

UNIT - V

9. a) Illustrate how lost segments are handled in Transmission control protocol. CO5 (07)
- b) Explain resolution and its types. CO5 (06)
- c) Elaborate on the different services offered by the e-mail user agent. CO5 (07)
10. a) Illustrate three-way handshaking in Transmission control protocol with a neat diagram. CO5 (07)
- b) Describe DNS with suitable examples and explain the following: CO5 (06)
- i) Generic Domain
 - ii) Country Domain.
- c) Illustrate with an example, how DNS can support an e-mail program to find the IP address of an e-mail recipient. CO5 (07)
